

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2024–2025
FULL MARKS: 75

CSE 4513: Software Engineering and Object-Oriented Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Although Line of Code (LOC) is the simplest metric for software size estimation, it is not widely used in practice. Do you agree? Justify your answer.

5
(CO1)
(PO1)

Solution:

- **Different complexity levels:** LOC is not consistent because all lines of code do not represent the same level of difficulty. Some lines are simple statements, while others involve complex logic, making them harder to design, implement, and test.
- **Developer dependency:** LOC varies from person to person, as different developers may write more or fewer lines to implement the same functionality based on their coding style, experience, and familiarity with the programming language.

- b) As a software engineer in a software company, you are assigned to develop an online payment system called VecnaPay. The team writes short requirements on cards. They discuss the details with stakeholders in meetings. Acceptance tests are used to check whether the features work as expected. Before finalizing the requirements document, you review the requirements to make sure they match stakeholder needs, requirements should not conflict with each other, include all necessary system functions such as payment validation and refund handling, are feasible to implement, and can be tested.

6 + 4
(CO1)
(PO1)

- i. Explain how the 3C approach can be used to write user stories for the VecnaPay system.

Solution:

3C approach for VecnaPay user stories:

- **Card:** Write short user stories on cards describing who the user is, what they want, and why.
- **Conversation:** Discuss the details of each story with stakeholders to clarify rules and constraints.
- **Confirmation:** Define acceptance tests to verify that the implemented features work as expected.

- ii. From the scenario, identify and explain any two checks that should be performed on the requirements to ensure good quality.

Solution:

- Validity
- Consistency
- Completeness

- Realism
- Verifiability

Any two of the above checks would be acceptable, provided they are properly justified from the scenario.

- c) Consider a software project with the following parameters: EI = 45, EO = 30, EQ = 25, ILF = 08, and EIF = 05. Assume all weighting factors are average. In addition, the system requires moderate performance, significant end-user efficiency, average distributed data processing, moderate data communication, and significant transaction rates. All other General System Characteristics (GSCs) are essential. Compute the Function Points of the system using Function Point Analysis.

8
(CO2)
(PO2)

Solution:

Using average weighting factors, the Unadjusted Function Points (UFP) of the system is computed as:

$$UFP = 545$$

From the General System Characteristics (GSCs), the Total Degree of Influence (TDI) is:

$$TDI = 60$$

Case 1: Using $VAF = (0.01 \times TDI) + 0.65$

$$VAF = 0.65 + 0.60 = 1.25$$

$$FP = 545 \times 1.25 = 681.25 \approx 681$$

Case 2: Using $VAF = (0.01 \times TDI) + 0.065$

$$VAF = 0.065 + 0.60 = 0.665$$

$$FP = 545 \times 0.665 = 362.43 \approx 362$$

Marking note: If students use either 0.65 or 0.065 in the Value Adjustment Factor formula, answers will be accepted and full marks will be awarded accordingly. If students are unable to recall the exact numerical values for terms such as average, moderate, significant, or essential in GSC ratings but demonstrate the correct method and reasoning, partial marks will be awarded.

2. The Hawkins Lab, located in the city of Hawkins, is developing a software system to monitor and analyze the city because stranger things are happening in different areas. Due to unclear and frequently changing requirements, the team follows Agile Scrum. In the early phase of the project, a prototype is created only to understand user requirements, and it will not be part of the final system. The team works in short sprints, tracks progress using Scrum artifacts, and places strong emphasis on both functional and non-functional quality before releasing features.

- a) Identify the most suitable prototyping model for the given scenario and justify your answer with appropriate reasoning.

5
(CO1)
(PO1)

Solution:

Most suitable prototyping model: Throwaway (Rapid) Prototyping

Justification: The prototype is developed only to understand unclear and frequently changing requirements and is discarded before building the final system.

- b) At the end of each sprint, the Hawkins Lab Scrum team conducts a Sprint Retrospective. Describe why the Sprint Retrospective is conducted and what the team aims to achieve through it.

5
(CO1)
(PO1)

Solution:

The Sprint Retrospective is conducted at the end of each sprint to help the Hawkins Lab Scrum Team reflect on how the sprint was executed.

During the retrospective, the team inspects what went well and what did not in terms of people, processes, and tools. The main aim is to identify areas for improvement and decide concrete actions, such as what to **start doing, stop doing, or continue doing**. Through this, the team seeks continuous improvement in collaboration, efficiency, and overall effectiveness in future sprints.

- c) The Hawkins Lab system should meet strict non-functional requirements such as reliability, security, and performance. Explain the importance of the “Definition of Done”, particularly in ensuring that non-functional requirements are satisfied.

5
(CO1)
(PO1)

Solution:

Definition of Done (DoD) is a shared agreement in the Scrum Team that defines when work is considered complete.

It is particularly important for **non-functional requirements (NFRs)** because NFRs are often vague and difficult to measure. DoD translates them into **clear and measurable criteria**, such as performance thresholds, security validations, and reliability tests. This ensures NFRs are consistently verified and not overlooked before a feature is marked as done.

- d) During a sprint, the Scrum team at Hawkins Lab notices that the actual burndown chart temporarily increases on some days instead of always decreasing. Explain the probable reasons for this situation and discuss whether it necessarily indicates a problem in the Scrum process.

6
(CO1)
(PO1)

Solution:

An increase in the burndown chart during a sprint usually indicates that **more work was added or that existing tasks were re-estimated**.

This can occur when new tasks are discovered, initial estimates were inaccurate, or technical challenges and dependencies increase the remaining effort. **Such an increase does not necessarily indicate a problem in the Scrum process, as it often reflects improved understanding and transparency.** However, frequent increases caused by uncontrolled scope changes or poor sprint planning may indicate issues that should be addressed in the Sprint Retrospective.

- e) While planning the sprint, a team member suggests converting story points into hours for easier estimation. Explain whether this practice is appropriate in Agile Scrum.

5
(CO1)
(PO1)

Solution:

Converting story points into hours is not appropriate in Agile Scrum because time varies from one person to another, while story points aim to measure the effort and amount of work required.

Two team members may need different amounts of time to complete the same task due to experience or skill level, but the overall effort and complexity remain the same. Story points focus on this relative effort, including complexity and uncertainty, rather than exact time. Converting them into hours defeats this purpose and leads to inaccurate and misleading estimates.

3. a) “General requirements such as friendliness should be avoided.” Discuss this statement in

5
(CO1)
(PO1)

the context of software requirements.

Solution:

General requirements such as “friendliness” should be avoided because they are vague, subjective, and not testable. Different stakeholders may interpret them differently, and developers cannot clearly determine how to implement or verify them. Such requirements should be replaced with specific and measurable statements.

- b) A newly deployed university portal is difficult for first-time users to understand, responds slowly during common tasks, and crashes when users make small input mistakes. From the user interface evaluation attributes, identify three relevant attributes that are not being satisfied and justify your choices.

6
(CO2)
(PO2)

Solution:

- **Learnability:** The portal is difficult for first-time users to understand, indicating that new users cannot become productive easily.
- **Speed of operation:** The system responds slowly during common tasks, showing poor responsiveness to users work practices.
- **Robustness:** The system crashes due to small input mistakes, meaning it is not tolerant of user errors.

- c) Write short notes on the following topics. Each note should include a brief explanation and one relevant example.

5 × 3
(CO1)
(PO1)

- YAGNI principle
- Mean Time to Failure (MTTF)
- KISS principle

Solution:

- **YAGNI (You Aren’t Gonna Need It):** This principle states that features should be implemented only when they are actually needed, not based on future assumptions. *Example:* Avoid adding a refund analytics module until there is a clear requirement for it.
- **Mean Time to Failure (MTTF):** MTTF is the average time a system operates before it fails, commonly used to measure reliability of non-repairable systems. *Example:* If a sensor-based system runs for an average of 500 hours before failing, its MTTF is 500 hours.
- **KISS (Keep It Simple, Stupid):** This principle emphasizes designing systems as simply as possible to reduce errors and improve maintainability. *Example:* Using a straightforward payment workflow instead of an overly complex multi-step process.

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	5	
1.b	10	
1.e	8	
2.a	5	
2.b	5	
2.c	5	
2.d	6	
2.e	5	
3.a	5	
3.b	6	
3.c	15	
Total	75	